

GROTON-NEW LONDON, CT, USA

WMO: 725046

Lat: 41.328N

Lon: 72.049W

Elev: 10

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 14707

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.0	13.7	-8.5	3.5	12.3	-4.0	4.5	17.6	28.5	27.5	26.1	29.7	11.6	310	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	84.4	72.6	81.7	71.8	79.5	70.8	75.7	80.9	74.4	78.9	73.1	77.2	9.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.0	126.7	78.6	72.8	121.7	77.1	71.9	117.8	76.0	39.1	80.7	37.8	79.1	36.7	77.4	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.5	20.8	18.7	DB	2.8	90.2	5.5	3.7	-1.1	92.8	-4.3	94.9	-7.4	97.0	-11.4	99.7	(4)
(5)			WB	1.3	78.1	5.1	1.9	-2.4	79.4	-5.4	80.5	-8.2	81.5	-11.9	82.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	30.3	31.9	38.2	47.6	56.6	65.5	71.8	71.4	65.2	54.7	45.2	36.2
	DBStd	16.08	9.59	8.16	7.65	6.21	5.83	5.49	4.48	4.56	5.78	7.52	7.64	8.30
	HDD50	2268	611	506	370	116	9	0	0	0	0	40	183	433
	HDD65	5572	1075	926	831	522	267	59	3	5	68	328	595	893
	CDD50	2756	1	0	4	44	215	466	676	665	457	185	38	5
	CDD65	585	0	0	0	0	7	75	214	205	75	9	0	0
	CDH74	2628	0	0	0	8	34	298	1059	998	219	12	0	0
	CDH80	447	0	0	0	0	3	51	213	159	19	2	0	0
Wind	WSAvg	8.8	10.1	9.9	10.3	9.5	8.1	7.1	6.7	6.8	8.0	9.3	9.6	9.9
Precipitation	PrecAvg	47.4	3.9	3.6	4.6	4.2	3.8	3.4	3.3	4.0	4.1	3.7	4.5	4.5
	PrecMax	64.5	12.2	6.7	9.2	10.4	7.7	14.0	7.7	11.6	9.3	8.1	10.9	9.7
	PrecMin	31.3	0.7	0.6	0.7	1.2	0.8	0.1	0.4	0.7	0.6	0.3	0.6	1.2
	PrecStd	6.9	2.1	1.5	1.9	2.0	1.8	2.7	1.8	2.4	2.0	1.9	2.4	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.6	54.3	61.1	73.0	78.6	85.2	89.6	87.6	82.5	75.0	65.0	59.3
		MCWB	51.6	48.3	50.8	57.4	65.7	71.2	74.8	74.8	70.8	67.0	59.2	56.3
	2%	DB	50.6	49.7	55.2	65.5	73.0	81.0	84.9	84.0	78.9	71.4	62.7	55.7
		MCWB	48.0	45.3	48.1	53.8	63.0	69.9	73.0	72.9	69.4	66.4	58.4	53.7
	5%	DB	47.6	46.7	52.1	60.8	69.8	77.2	81.9	81.4	76.4	69.1	60.4	53.0
		MCWB	44.7	43.2	46.6	50.8	60.9	67.8	72.1	72.4	69.1	64.5	56.6	51.0
	10%	DB	44.7	44.1	49.2	57.3	66.2	74.4	79.5	79.2	74.2	66.5	57.8	49.9
		MCWB	41.5	40.4	44.5	50.1	58.8	66.6	71.1	71.5	68.4	62.4	54.3	46.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	50.7	54.5	61.1	68.6	74.5	78.2	78.0	75.4	70.2	62.4	57.6
		MCDB	53.7	52.1	57.1	67.1	75.1	81.6	86.1	84.3	79.7	72.7	64.0	58.3
	2%	WB	48.5	46.9	50.9	56.9	65.5	72.3	75.7	75.9	73.1	67.7	60.1	54.0
		MCDB	49.9	48.7	53.3	61.9	70.6	78.0	81.4	80.7	75.8	70.2	61.7	54.8
	5%	WB	45.6	44.2	48.0	54.2	63.1	70.2	74.3	74.7	71.6	65.8	57.5	51.3
		MCDB	47.0	46.0	50.6	57.7	67.7	74.8	79.2	78.9	74.2	68.4	59.4	52.6
	10%	WB	41.7	41.0	45.6	51.9	60.8	68.4	72.9	73.4	70.2	63.4	55.0	47.5
		MCDB	44.3	43.6	48.3	55.3	64.7	72.5	77.5	77.2	72.9	65.8	57.2	49.3
Mean Daily Temperature Range	5% DB	MDBR	14.2	14.7	14.7	15.6	14.9	14.1	13.5	13.9	15.2	15.9	15.5	13.9
		MCDBR	16.6	17.5	17.8	20.9	19.8	18.1	16.0	15.0	15.9	16.0	15.7	15.6
		MCWBR	15.1	14.5	13.0	12.0	11.3	10.0	8.0	8.1	10.0	12.0	13.6	16.1
	5% WB	MCDBR	16.1	16.8	15.7	16.5	16.6	15.4	13.1	11.8	12.1	13.4	13.9	15.5
		MCWBR	15.9	15.6	13.4	11.6	10.9	9.8	7.7	7.1	9.2	11.6	14.1	16.6
Clear-Sky Solar Irradiance	taub	0.306	0.315	0.337	0.376	0.424	0.465	0.488	0.453	0.382	0.345	0.316	0.304	
	taud	2.496	2.472	2.454	2.375	2.270	2.196	2.143	2.233	2.406	2.519	2.554	2.552	
	Ebn at Noon	270	285	290	284	271	259	252	258	271	271	265	262	
	Edh at Noon	23	28	32	37	42	45	48	42	33	27	22	21	
All-Sky Solar Radiation	RadAvg	591	869	1191	1545	1719	1876	1912	1712	1403	946	651	495	
	RadStd	48	82	91	165	168	188	103	96	107	88	49	46	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.35	N/A	N/A	+1.51	N/A	N/A	N/A	N/A	+243	N/A
(47) Regional (76 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+184	+99

Nomenclature: See separate page